

Knowledge-Enhanced Attributed Multi-Task Learning for Medicine Recommendation

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Medicine recommendation systems target to recommend a set of medicines given a set of symptoms which play a crucial role in assisting doctors in their daily clinics. Existing approaches are either rule-based or supervised. However, the former heavily relies on expert labeling, which is time-consuming and costly to collect, and the latter suffers from the data sparse problem. To automate medicine recommendation on sparse data, we propose MedRec, which introduces two graphs in modeling: (1) a knowledge graph connecting diseases, medicines, symptoms, and examinations; (2) an attribute graph connecting medicines via shared attributes and molecular structures. These two graphs enhance the connectivity between symptoms and medicines, which thus alleviate the data sparse problem. By learning the interrelationship between diseases, medicines, symptoms and examinations and the inner relationship within medicine, we can acquire unified embedding representations of symptoms and medicines which can be used in medicine recommendation. The experimental results show that the proposed model outperforms state-of-the-art methods. In addition, we find that these two tasks: learning graph representation and medical recommendation can benefit each other.

CCS Concepts: • **Information systems → Recommender systems;**

Additional Key Words and Phrases: Medicine recommendation, knowledge graph embedding, multi-task, graph neural network

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1 INTRODUCTION

Medicine recommendation play a crucial role in assisting doctors to make accurate medicine prescriptions [11, 18]. Given a set of symptoms, medicine recommendation systems target to predict a set of medicines which can treat these symptoms without causing harmful medicine-medicine interactions. Different from traditional recommendation tasks, which model the single relationship between one user and one item, the medicine recommendation needs to jointly consider a set of symptoms and predict a set of suitable medicines.

Due to the practical value in clinics, medicine recommendation has attracted increasing research interest. There are two major research lines: (1) rule-based and (2) supervised models. Rule-based methods like [6] heavily rely on manual crafted rules from experienced doctors, which are both time-consuming and costly. Furthermore, manual crafted rules might only be effective for common symptom combinations, as to complicated cases, like patients with an excessive number of symptoms or rarely symptom combinations, these rules may fail to generate recommendations.

Supervised learning algorithms, such as Multi-Instance Multi-label learning [65], are trained to output multiple labels as medicine recommendations. The quality of supervised methods heavily depends on the quality of the available data. However, in practice, for long-tailed symptoms and medicines, there may be only several mentions in the training set, making it extremely difficult to train an accurate recommendation model. In other word, medicine recommendation suffers from the data sparsity problem. Take the Medicine dataset[1] for example, this dataset includes 4,128 symptoms and 400 medicines, but there are only 5,031 pairs of these symptoms and medicines. Considering all possible combinations ($4{,}128 \times 400$), this is quite a sparse dataset.

The input of the recommendation is a symptom set and a candidate medicine set, it does not contain enough information of connectivity between symptoms and medicines. To enhance the connectivity between symptoms and medicines, we attempt to incorporate two graphs in modeling: (1) Medical knowledge graph and (2) Medicine attribute graph. The first solution is to map the symptoms and medicines into a medical knowledge graph, to obtain the rich representation of symptoms and medicines. In the context of this article, the medical knowledge graph consists of four kinds of nodes: medicine, symptom, disease, and physical examination. If two nodes are related, they are connected via an edge. For example, a disease is connected to its typical symptoms and the medicines which can cure this disease. By incorporating the knowledge graph, the symptoms and medicines can be indirectly related if they share the same disease. In this manner, the medical knowledge graph can enhance the connectivity between symptoms and medicines.

However, the medical knowledge graph is usually built by mining medical documents, like textbook and EHR [22]. These medical documents may not cover all medical knowledge, thus the generated knowledge graph could be incomplete. For example, the edge between *cold* and *paracetamol* is missing in the medical knowledge graph in Figure 1. In addition, the medical knowledge graph only represents the explicit relations between symptoms, medicines, diseases, and physical examinations, but it lacks the medical facts of medicine, such as the molecular structure and drug-drug interactions. Therefore, we further introduce an attribute graph to model the inner

[1]The detailed description of this dataset can be found in Section 4.1.

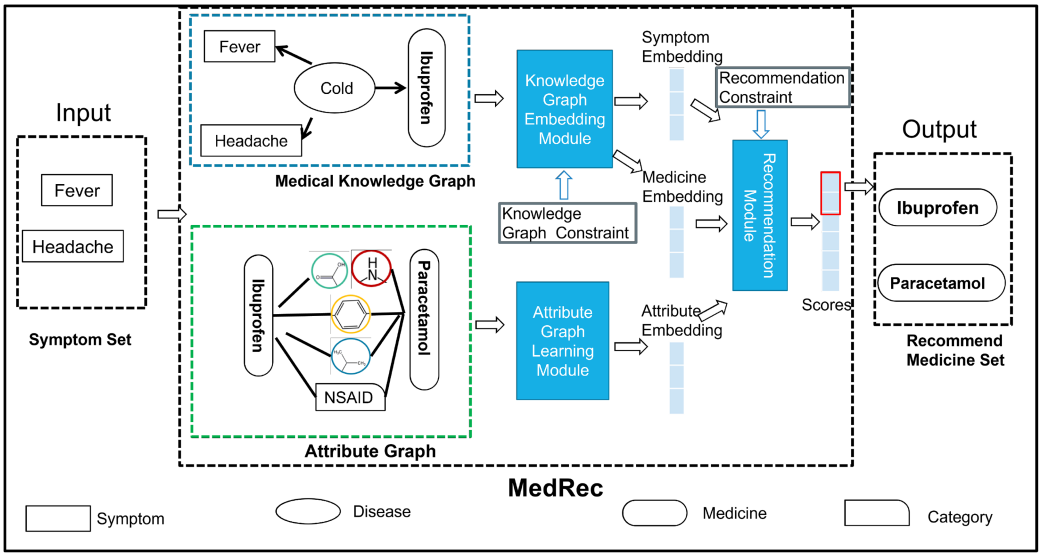


Fig. 1. Given the Symptom set, our proposed MedRec gives the recommended medicines based on the medical knowledge graph and attribute graph. Our method recommends "ibuprofen" according to the medical knowledge graph. Even though the relation between "paracetamol" and "cold" is missing in the medical knowledge graph, our method can recommend "paracetamol" according to the attribute graph.

relationship among medicines. Here, two medicines are connected if they share the same category or submolecular structure.

Take Figure 1 as an example, given a patient with symptoms *fever* and *headache*, through the knowledge graph, we can infer that this patient catches a cold and recommend *ibuprofen* which can ease the fever and headache caused by the common cold. In addition, we leverage the medicine attribute graph to find *paracetamol*, even through the edge between *cold* and *paracetamol* is missing in the medical knowledge graph. This is because both *ibuprofen* and *paracetamol* belong to the category of **Non-Steroidal Anti-Inflammatory Drug** (**NSAID**) and have three sub-molecular structures in common. Figure 1 provides an example of medicine recommendation with the joint efforts of medical knowledge graph and medicine attribute graph.

To model both medical knowledge graph and medicine attribute graph, we propose MedRec, a multi-task learning framework for medicine recommendation. MedRec conducts recommendation with the following three steps: (1) On the medical knowledge graph, MedRec first learns the entity and relation's embedding via multi-relational **Graph Convolutional Network** (**GCN**), and optimizes the model with the objective function of the link prediction task. In this manner, we can acquire the embedding of symptoms and medicines. To represent a set of symptoms, we fuse the embedding of each symptom with attention mechanism and generate the syndrome representation; (2) To learn the representation of a medicine, in addition to the embedding obtained from the medical knowledge graph, we also mine the medicine attribute graph with GCN to acquire another embedding. The embeddings obtained from the medical knowledge graph and medicine attribute graph are concatenated together to form the final representation of a medicine; (3) We learn the interaction of syndrome and medicine to produce the prediction scores. To merge the above three steps in one unified framework, we propose a multi-task framework to handle the training process of the graph embedding and medicine recommendation tasks.

We conduct experiments of MedRec on two datasets, one public and one private. The experimental results demonstrate the effectiveness of MedRec and show that the two tasks (graph embedding learning and medicine recommendation) are highly correlated and can benefit each other.

The main contributions of this article are summarized as follows:

— We introduce the medical knowledge graph, which can model the interrelationship between symptoms, medicines, diseases, and physical examinations.
— We introduce the medicine attribute graph, which models the inner relationship within medicines by exploiting the category information and molecular structures.
— By learning the embedding representation from the above two graphs, we can alleviate the data sparse problem.
— We apply a novel multi-task framework for graph embedding learning and medicine recommendation, these two tasks are correlated with each other and can benefit the performance of each other.

2 RELATED WORK

In this section, we review related work about the **graph neural network** (**GNN**) based recommendation system in Section 2.1, the knowledge graph embedding in Section 2.2, molecule structure in Section 2.3, multi-task framework for recommendation in Section 2.4 and medicine recommendations in Section 2.5.

2.1 GNN-Based Recommendation Systems

Due to the effectiveness and superior performance in many applications, GNNs [33, 54, 55, 57] has also obtained increasing interest in recommender systems [3, 5, 9, 34, 38, 68]. UGrec [46] models user and item interactions as a graph network, defines sequential recommendation paths from users' purchased histories, and further aggregates different paths using an attention mechanism. Reference [56] uses GNN for session-based recommendation by capturing more complex relationships between items in a sequence. Each session is represented as the composition of the long-term preference and short-term interests within a session using an attention network. GCE-GNN [51] employs a session-aware attention mechanism to recursively incorporate the neighbors' embeddings of each node in the global graph. GraReg [66] is a graph-based regularization approach to serve as a counterpart of the L2 regularization for embedding layers. RippleNet [43, 44] considers the knowledge graph as the source of side information and addresses the limitations of existing embedding-based and path-based methods for KG-aware recommendations. MBGCN [17] learns behavior strength by user-item propagation layer and captures the behavior semantics by item-item propagation layer, which exploits the semantics of multi-behavior data. In addition, due to the effectiveness in representation learning, GNNs are widely exploited to model the interaction for recommendation. For instance, [49] proposes a neural graph collaborative filtering method, where the collaborative signal conveyed by the edges and high-order connectivity is explicitly modeled and injected into the user and item embedding. MMGCN [52] constructs the model-specific bipartite graph to model the user preference in multiple modalities, and represents the user preference in each modality with her/his directly and indirectly connected neighbors.

However, as can be seen, these models fail to capture the additional commonsense information, but only focus on the relationship between user and item. In our work, we utilize the medical knowledge graph and medicine attributes to help improve the recommendation performance.

2.2 Knowledge Graph Embedding

Knowledge graph embedding is to project each entity and relation in the knowledge graph into a low-dimensional space. There are mainly two research lines for knowledge graph embedding: translation distance-based methods and semantic matching methods.

Translation distance-based methods, such as TransE [2], TransR [24], build entity and relation embedding by regarding a relation as a translation from head entity to tail entity. RotatE [40] extends the computation into the complex domain and is currently the state-of-the-art model in this category. OPTransE [69] takes relation paths into consideration and projects the head entity and the tail entity of each relation into different spaces to guarantee the order of relations in the path. RatE [16] uses a relation-adaptive translation function built upon a novel weighted product in complex space, where the weights are relation-specific and independent to embedding size. Moreover, some methods incorporate the ontology knowledge graph, which constitute semantic meta-relations of abstract concept in to knowledge graph embedding learning. JOIE [12] and BioJOIE [13] jointly embed the instance-view graph and the ontology-view graph by leveraging triples in both graphs and cross-view links that connect the two graphs, capturing latent features from both views in KBs.

Semantic matching models include DistMult [58] and ComplEx [41]. KGNN [23] learns the representation of drugs in the knowledge graph and applies it on the drug-drug interaction prediction. RGHAT [67] uses relation-level and entity-level attention to highlight the importance of different neighboring entities and utilizes the neighborhood information of an entity. DREM [1] is a dynamic relation embedding model that jointly learns embedding representations for entities/relationships and learns both static and dynamic relationships. AprilE [28] employs triple-level self-attention and pseudo residual connection to model relational patterns.

However, the above knowledge graph embedding methods mainly focus on modeling individual triples and ignore the knowledge graph structure which cannot take full advantage of the context of the neighbors. It inspires us to use GNNs to capture the global information of the knowledge graph [20, 32].There are several methods that apply heterogeneous GNN on knowledge graph embedding learning. MEDTO [14] applies hyperbolic graphs convolution network and heterogeneous graph network on ontology graph to learn the embedding of the ontology and completes a medical data to the ontology matching task. HAN [50] utilizes the hierarchical attention, including node-level attention and semantic-level attention. HGT [15] further uses node- and edge-type dependent parameters to characterize the heterogeneous attention over each edge. NARS [63] computes neighbor averaged features for random subsets of relation types and combines the features for the down streaming task.

2.3 Molecule Structure in Drug Discovery

The molecule structure contains information about the property of the medicine, such as absorption, distribution, metabolism, and excretion property. **Extended-Connectivity Fingerprints** (**ECFP**) encodes the neighboring environment of heavy atoms in a compound with a hashed integer identifier [36], with each unique identifier corresponding to a unique compound substructure, called fingerprint. The molecular property learning can be categorized into two main groups: (1) molecular graph and (2) **Simplified Molecular-Input Line-Entry** (**SMILES**) notation [53]. Previous research focused on transforming the graph structure of molecules to a fixed size of feature descriptors, then these descriptors can be easily used by existing machine learning algorithms. In the first graph based method, each molecule is represented as a graph associated with different nodes interconnected by bond edges. Some approaches employ GNNs to learn molecular representations [29, 30]. For example, MGCN [29] is a hierarchical GNN, which includes an embedding

layer, a Radial Basis Function Layer and an interaction layer to learn molecular representations. Chemi-Net [25] applies deep neural network to the molecule graph and predicts the properties. Molecules can be generated as sequential objects using SMILES notation [53]. Then, the sequential object can be adopted as language models to learn their representation [47, 64].

However, SMILES strings do not preserve the full connectivity information, as molecules are more represented as undirected graphs, with finite sets of vertex and edge types. In this article, MedRec is proposed to capture the molecular global information (sub-molecular) during the recommendation.

2.4 Multi-Task Framework for Recommendation

Multi-task learning [26, 27, 35, 37] is a learning paradigm in deep learning where it aims to leverage the information contained in multiple related tasks to help improve the generalization performance of all the tasks. All of these learning tasks are assumed to be related to each other, and it is found that learning these tasks jointly can lead to performance improvement compared with learning them individually. Hard parameter sharing [4] and soft parameter sharing [8, 60] are two most basic and commonly used in multitask learning structure. Hard parameter sharing is applied by sharing the hidden layers between all the tasks, except for several task-specific output layers. On the other hand, in soft parameter sharing, each task has its own model parameters, and there are constrains between the network parameters, which is inspired by regularization techniques.

Besides hard parameter sharing and soft parameter sharing, there are some recommender systems applied to multitask models with efficient shared learning mechanism, which aims at improving the accuracy of recommendation systems. MKR [45] aims at utilizing the connection between the recommendation system and knowledge graph to help improve their performance. DUPN [31] learns universal user representations across multiple tasks, including click-through rate prediction task, learning to rank task, and so on. User representations are shared by different tasks. NMTR [10] learns recommendation systems for the cascading relationship among different types of behavior, where the optimization of each behavior is treated as a task. [7] proposes a model that improves both the prediction accuracy and explainability of recommendation by fully exploiting the correlations between the recommendation task and the explanation task. AAMR [48] is a generic Attribute-Aware Multi-task Recommendation framework for rating prediction and social prediction, which learns representations for users and items by preserving both rating data and social data and attribute information, to conduct both rating prediction and trust relationship prediction tasks. MTRec [21] relies on self-attention mechanism to learn the semantics of meta-paths in a heterogeneous information network and jointly optimizes the tasks of both recommendation and link prediction.

Our work is an asymmetric multi-task learning framework. We utilize the connection between the recommendation task and knowledge graph learning task. The model is trained in an iterative manner.

2.5 Medicine Recommendation

Medicine recommendation systems play a crucial role in assisting doctors to make accurate medicine prescriptions. The medicine recommendation based on the prescription contains three stages: (1) frequency statistic and data mining methods, including association analysis, simple clustering; (2) topic models, which compute the conditional probability of co-occurred symptom and medicine words to capture the relations among symptoms and medicines [61]; (3) graph-based methods. For example, SMR [11] proposes a novel method for modeling a patient based on the learned graph embeddings and makes a safe recommendation by minimizing the potential adverse drug reactions. In SMGCN [18], the symptoms and medicine are trained to obtain a

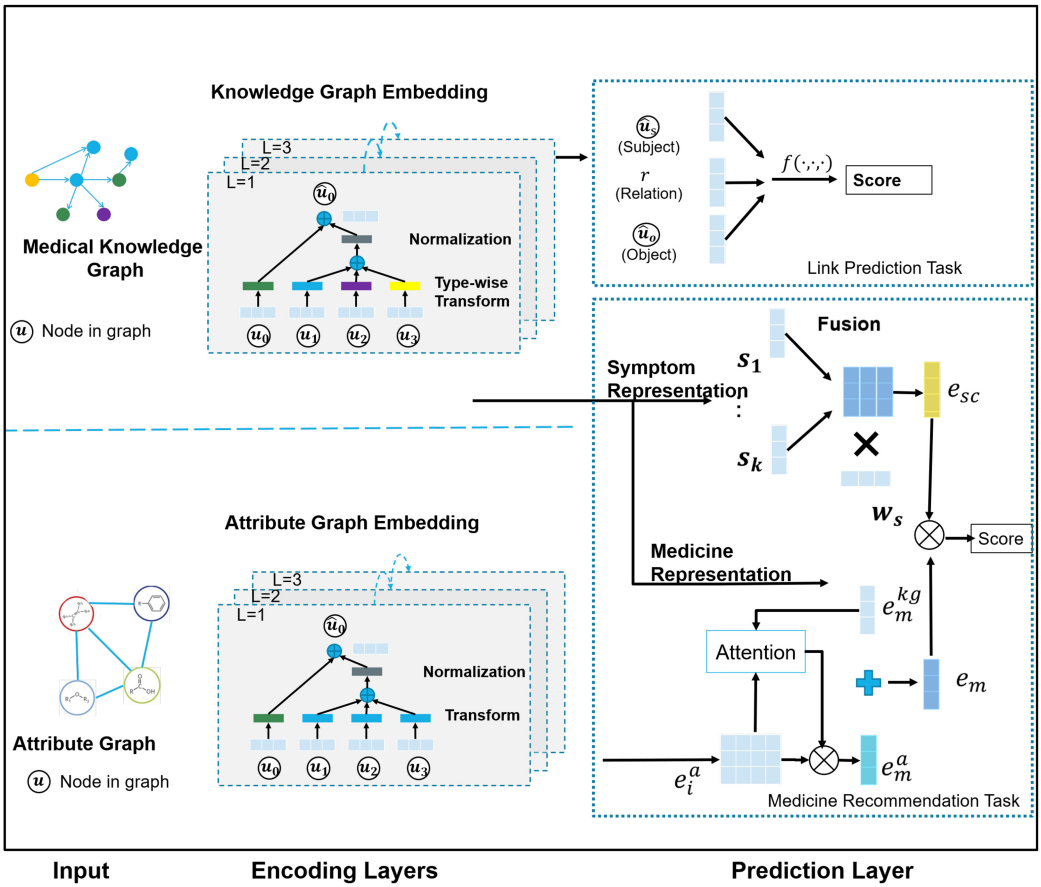


Fig. 2. The architecture of the proposed MedRec which includes three components: (1) the knowledge graph learns the representations of medicines, symptoms, diseases and physical examinations; (2) the attribute graph learns another representation of medicines; (3) the recommendation model combines acquired graph embedding to recommend suitable medicines for symptoms.

predictive model that outputs multiple labels as medicine recommendations. SafeDrug [59] uses a global **message passing neural network** (**MPNN**) module and a local bipartite learning module to fully encode the connectivity and functionality of drug molecules, therefore capturing both global and local information during the recommendation.

Despite their initial success, these existing works still have the following limitations. Other important data such as medicine category information and molecule structures can help augment medicine recommendation but have not been utilized by existing models. This article proposes a multi-task framework that leverages medicine's attributes and interaction with the symptoms effectively.

3 METHODOLOGY

In this section, we introduce the proposed MedRec in detail. The framework of MedRec is illustrated in Figure 2. We introduce two graphs in medical recommendation: (1) the knowledge graph that exploits the interrelationship between medicines, symptoms, diseases, and physical examination; (2) the attribute graph that exploits the inner relationship within medicines.

Table 1. Notations and Their Definitions

Notations	Definitions
$\mathcal{G}^{kg}$	Medical knowledge graph
$\mathcal{G}^{a}$	Attribute graph
$\mathcal{E}^{kg}$	Set of entities in the medical knowledge graph
$\mathcal{R}^{kg}$	Set of relations in the medical knowledge graph
$\mathcal{T}^{kg}$	Set of medical knowledge graph triples
$\mathcal{E}^{a}$	Set of entities in the medicine attribute graph
$\mathcal{R}^{a}$	Set of relations in the medicine attribute graph
$\mathcal{T}^{a}$	Set of medicine attribute graph triples
sc	Set of symptoms in a prescription
mc	Set of medicines in a prescription
d	Dimension of the vector
M	Collection of all medicines
S	Collection of all symptoms
$\mathcal{N}_i^r$	Neighbors of node i with relation r in the knowledge graph
$\mathcal{N}_i$	Neighbors of node i in the attribute graph
w_s	Weight of symptom regard to the frequencies
e_m^{kg}	Medicine representation from medical knowledge graph
e_m^{a}	Medicine representation from medicine attribute graph
e_i^{a}	Attribute representation from medicine attribute graph
e_s	Symptom representation from medical knowledge graph
e_M	Medicine representation matrix
e_{sc}	Representation of set of symptoms

In addition to the categorical attributes, the attribute graph also considers the molecular structure of medicines and connects medicines that share similar molecular structure, which is novel. The two graphs are unified by a multi-task learning framework to improve recommendation.

3.1 Problem Definition

We formulate the joint learning of graph embedding and medicine recommendation as follows. The input includes the knowledge graph $\mathcal{G}^{kg} = \{\mathcal{E}^{kg}, \mathcal{R}^{kg}, \mathcal{T}^{kg}\}$, attribute graph $\mathcal{G}^{a} = \{\mathcal{E}^{a}, \mathcal{R}^{a}, \mathcal{T}^{a}\}$ and a set of prescription, where $\mathcal{E}^{kg}$ is the set of entities with four types: symptoms, medicine, disease and physical examinations, $\mathcal{E}^{a}$ is the set of attribute entities, $\mathcal{R}^{kg}/\mathcal{R}^{a}$ is the set of relations, $\mathcal{T}^{kg}/\mathcal{T}^{a}$ is the set of triples. Each prescription consists of a symptom set $sc = \{s_1, s_2, \dots\}$ and medicine set $mc = \{m_1, m_2, \dots\}$, and symptom and medicine can be mapped into entities in the knowledge graph.

For the medicine recommendation task, our goal is to complete an N-dimensional probability vector with a prediction function $y = \mathcal{F}(mc|\Theta, \mathcal{G}^{kg}, \mathcal{G}^{a}, sc)$, where the value of dimension i represents the probability that medicine i can cure sc, Θ is the parameters of the model. For the knowledge graph embedding task, our goal is to learn representations for entities and relations in the knowledge graph. Table 1 summarizes the key notations used in the rest of this article.

3.2 Graph Construction

MedRec contains two kinds of graphs: the medical knowledge graph and the medicine attribute graph. In this subsection, we introduce how to construct the medical knowledge graph and medicine attribute graph.

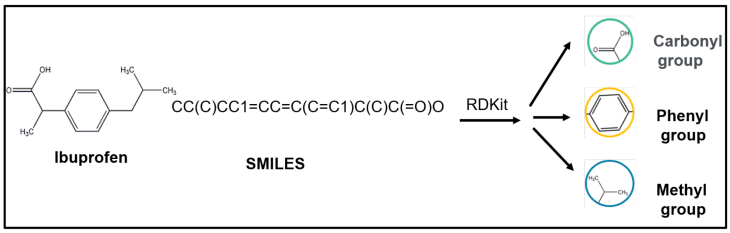


Fig. 3. The building steps of medicine attribute graph. Given the medicine Ibuprofen, (1) we first acquire the SMILES string of this medicine which is a textual representation of the molecular structure; (2) we use the RDKit to break down the SMILES string into three sub-strings, each of which represents a sub molecular structure.

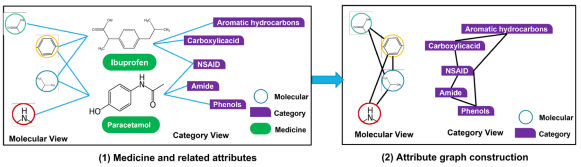


Fig. 4. The building steps of medicine attribute graph. Find the relation between medicines and attribute and construct the attribute graphs.

3.2.1 Medical Knowledge Graph. The medical knowledge graph we use here contains four kinds of entities and several relations between them. For Medicine dataset, The entity types include symptom, medicine, disease, and physical examination. The edges between the entities show the relations between the entities, such as <"aspirin","cure","headache">. For TCM dataset, we construct the knowledge graph based on the prescription set. The entity $\mathcal{E}$ is the combination of symptom set and medicine set. In addition, we define four different relations: "HerbSym","SymHerb", which describe the interaction between the pair of medicine set and symptom set, and "HerbHerb", "SymSym", which indicate the interaction within the symptom set or medicine set.

3.2.2 Medicine Attribute Graph Construction. Figures 3 and 4 illustrate how to build the medical attribute graph. Given the medicine Ibuprofen, we first acquire the SMILES string of this medicine which is a textual representation of its molecular structure; then, we use the RDKit[2] to break down the long SMILES string into three sub-strings: *Carbonyl group*, *Phenyl group*, and *Methyl group*. Each sub string represents a sub-molecular structure. After the segmentation, each molecule is decomposed into a set of sub-molecule structures, denoted as v_m. We also acquire the category information: *Aromatic hydrocarbons*, *Carboxylicacid*, and *NSAID*. If two sub-molecules/categories (nodes) share one medicine, they are connected by an edge in the attribute graph, as shown in Figure 4(2). In this manner, the two medicines *Ibuprofen* and *Paracetamol* can be indirectly related as they share the same categories and sub-molecular structures.

3.3 Graph Embedding Learning

3.3.1 Medical Knowledge Graph Embedding Learning. Graph embedding is to learn the continuous representation of entities and relations while preserving the structure and semantic information. GNNs encode neighbor information to take advantage of the context of the neighbors. We use R-GCN [39] as our graph convolution layer, but the framework is agnostic to the particular

[2]https://www.rdkit.org/.

choice of graph convolution layers. We define the following propagation model for calculating the forward-pass update:

$$h_i^{(l+1)} = \sigma\left(\sum_{r\in\mathcal{R}}\sum_{j\in\mathcal{N}_i^r}\frac{1}{|\mathcal{N}_i^r|}W_r^{(l)}h_j^{(l)} + W_0^{(l)}h_i^{(l)}\right). \tag{1}$$

After updating the representation of the entities and relations in the knowledge graph, denoted as $e_i = h_i^{(L)}$, where L is the number of graph convolution layers. We then rank the triples in the knowledge graph with score functions. In our experiment, we use RotatE score function [40] as the score function, which is known to perform well on the standard link prediction task. In RotatE, a triple is scored as

$$f(sub, rel, obj) = -||e_{sub} \circ e_{rel} - e_{obj}||^2, \tag{2}$$

where (sub, rel, obj) is the triple in the knowledge graph, $e_{sub}, e_{rel}, e_{obj}$ is the representation of the subject, relation and object. They are mapped into complex vector space $\mathbb{C}$, $|e_{rel}| = 1$, and $\circ$ denotes the Hadamard product.

We train the model with negative sampling. For each triple in the knowledge graph, we sample ω negative ones by randomly corrupting either the subject or the object of each positive triple. We optimize for cross-entropy loss to push the model to score positive triples higher than the negative ones:

$$\mathcal{L}_{KG} = -\frac{1}{(1+\omega)|\mathcal{E}|}\sum_{(sub,rel,obj,y)\in\mathcal{T}}(y\log(\sigma(f(sub,rel,obj))) + (1-y)\log(1-\sigma(f(sub,rel,obj)))), \tag{3}$$

where $\mathcal{T}$ contains the positive and negative triples, $y \in \{1, 0\}$ indicates whether it is the positive or negative triples, and σ sigmoid function.

3.3.2 Medicine Attribute Graph Embedding Learning. We treat attribute embedding learning as a representation learning problem on a homogeneous graph. The representation of each attribute and the relationship between them are learned by the GCN [20]:

$$h_i^{(l+1)} = \sigma\left(\sum_{j\in\mathcal{N}_i} W^{(l)}h_j^{(l)} + W_0^{(l)}h_i^{(l)}\right), \tag{4}$$

where $\sigma(x) = \frac{1}{\exp(-x)+1}$ is the sigmoid function.

Once the forward propagation of GCN layer has finished, through learning graph embedding on both medical knowledge graph and medicine attribute graph, we can obtain the feature vector of all nodes, which include single symptom representation e_s and medicine representation e_m^{kg} from medical knowledge graph, the attribute representation $e_i^a = h_i^{(L)}$ from attribute graph. Here, e_i^a denotes the embedding of either the category name or the sub-molecular structure.

3.4 Medicine Recommendation

3.4.1 Medicine Representation. The representation of the single medicine attribute can be aggregated as follows:

$$e_v^a = \sum_{k\in v_m} e_k^a, \tag{5}$$

where v_m denotes the set of attributes of the same type. In the context of this article, given a medicine, v_m refers to two attribute sets of this medicine: the set of category names and the set of sub molecular structures. In this manner, e_v^a can denote (1) the aggregated category embedding (2) the aggregated molecular embedding. To further merge these two aggregated embeddings, the

representation of medicine is calculated by Equation (6), which consider the varied importance of different attribute sets:

$$\hat{e}_v^a = \text{softmax}\left(\frac{e_m^{kg} \cdot e_v^{a\top}}{\sqrt{d}}\right) e_v^a, \tag{6}$$

where e_m^{kg} is the medicine representation from knowledge graph embedding and d is the dimension of the embedding. Then, the medicine embedding in the medicine attribute graph is represented as

$$e_m^a = \sum_{v=1}^{V} \hat{e}_v^a. \tag{7}$$

The final representation of medicine is the concatenation of two parts: one from the knowledge graph e_m^{kg} and the other is the medicine attribute graph e_m^a, denoted as

$$e_m = e_m^{kg} \oplus e_m^a, \tag{8}$$

which is a unified representation of a single medicine from both knowledge graph and attribute graph. We can construct the medicine representation matrix $e_M \in \mathbb{R}^{|M| \times d}$ where each row in e_M is e_m from Equation (7) and $|M|$ denotes the size of the medicine collection.

3.4.2 Symptom Set Representation. In order to acquire the symptom representation e_s from the knowledge graph, we use the weighted average to aggregate the symptom representations to obtain the representation of a single symptom set sc in a prescription:

$$e_{sc} = \sum_{s \in sc} \frac{w_s}{\sum_{s \in sc} w_s} e_s. \tag{9}$$

By analyzing the data statistics, we find that the frequency of symptoms is varying among different symptoms. For example, in 28,746 prescriptions of the training set of TCM dataset, 29 symptoms appear over 400 times, and the highest appears 3,055 times. However, half of the symptoms appears less than 15 times. Therefore, we design the weight of the symptom w_s to balance the contribution of symptoms with various frequencies:

$$w_s = \log\left(\frac{|T|}{freq(s) + 1}\right), \tag{10}$$

where T is the training set of the prescriptions. The more frequent symptoms appear in different train cases, the lower its weight is, since it indicates the symptom is general. This force our model to focus on rare symptoms instead of general symptoms like "headache".

3.4.3 Recommendation. Here, we use $score(sc, M)$ to characterize the ranking score in the medicine recommendation, where e_M is the embedding matrix for the medicine collection M and σ is the sigmoid function:

$$score(sc, M) = \sigma(e_{sc} \cdot e_M^{\top}), \tag{11}$$

Given the input symptom set sc, the ground truth set mc is represented as a multi-hot vector mc in dimension $|M|$, $score(sc, M)$ is the output probability vector for all medicine. We define the mean square loss between mc and $score(sc, M)$ as follows:

$$\mathcal{L}_{RS} = \sum_{i=1}^{|M|} (mc_i - score(sc, M)_i)^2. \tag{12}$$

ALGORITHM 1: The Multi-Task Training Process.

Input: knowledge graph $\mathcal{G}^{kg}$, symptoms-medicines pair (sc, mc), attribute graph $\mathcal{G}^{a}$;
Output: Prediction function $\mathcal{F}(mc|\Theta, \mathcal{G}^{kg}, \mathcal{G}^{a}, sc)$;

1: Initialize all parameters;
2: Calculate the weight of the symptoms w_s;
3: **for** number of training iteration **do**
4: Sample minibatch of true and false triples from knowledge graph $\mathcal{G}^{kg}$;
5: Apply graph convolution operation on knowledge graph $\mathcal{G}^{kg}$ according to Equation (1);
6: Calculate the score of the knowledge triples in Equation (2);
7: Update parameter of $\mathcal{F}$ by gradient descent in Equation (3);
8: **for** t steps **do**
9: Sample minibatch of symptoms-medicines pair (sc, mc);
10: Apply graph convolution operation on attribute graph $\mathcal{G}^{a}$ according to Equation (4);
11: Acquire the symptoms and medicine's knowledge graph representation from $\mathcal{G}^{kg}$;
12: Calculate the representation of the medicine attribute based on the attribute graph $\mathcal{G}^{a}$;
13: Generate the symptom set representation and medicine representation in Equations (8) and (9);
14: Calculate the score of the medicines in Equation (11);
15: Update the parameters of $\mathcal{F}$ by gradient descent in Equation (12);
16: **end for**
17: **end for**
return $\mathcal{F}(mc|\Theta, \mathcal{G}^{kg}, \mathcal{G}^{a}, sc)$;

3.5 Multi-Task Training

MedRec is trained in an end-to-end manner. Each training epoch consists of two stages: graph embedding learning and medicine recommendation, and the training algorithm is shown in Algorithm 1. The model is trained in an iterative manner. In each iteration, the parameters of medical knowledge graph are firstly optimized (lines 4–7), and then the parameters of the recommendation system are optimized (lines 9–15). In other words, the parameters of graph embedding and recommendation systems are mutual influenced during optimization which could be regarded as an end-to-end approach. We optimize the model with Adam [19] algorithm. We repeat the training on the recommendation task for t times after training on the graph embedding task. The choice of t will be discussed in Section 4.

The graph embedding learning provides rich information of symptoms and medicines, which can benefit the medicine recommendation task. On the other hand, since the parameters of the GNN in Section 3.3 are updated during the training of recommendation task, the pair of symptom set and medicine set can be regarded as the proximity constraint of the graph embedding learning task, which is the high-level interaction of entities.

4 EXPERIMENTS

In this section, we evaluate our proposed MedRec on two datasets TCM [61] and Medicine. The experiments are conducted to address the following questions.

RQ1: Could the proposed model outperform state-of-the-art approaches on two tasks: graph embedding learning and medicine recommendation? (in Section 4.5)

RQ2: What is the contribution of each proposed component to the overall recommendation performance, including medical knowledge graph, medicine attribute graph and the

Table 2. Statistics of the Evaluation Datasets for Knowledge Graph Embedding

Dataset	#Ent	#Rel	#Train	#Test
TCM	1,201	4	216,100	104,050
Medicine	21,478	37	186,434	46,624

Table 3. Statistics of the Evaluation Datasets for Recommendation

Dataset	#Sym	#Med	#Train	#Test
TCM	390	811	28,746	5,019
Medicine	4,128	400	17,576	4,395

recommendation model? Will learning graph embedding improve the performance of recommendation and will the optimization of recommendation improve the performance of graph embedding learning? (in Section 4.6)

RQ3: What is the performance w.r.t different hyper-parameter settings (e.g., embedding dimension d, depth of the neural network layers L, and training frequency of recommendation module t)? (in Section 4.7)

RQ4: How to explain the recommendation results? (in Section 4.8).

4.1 Datasets

The performance of the proposed MedRec is evaluated on two datasets, a public dataset TCM [61], and a private dataset Medicine, The statistics of these two datasets are summarized in Tables 2 and 3.

— **Medicine**. We collect the data and build the Medicine dataset ourselves. Medicine contains 21,478 entities, including four types: symptoms, medicines, diseases, and physical examinations and 37 kinds of relationships. Then we process the knowledge graph to obtain the pair of symptoms and medicine collection (sc, mc). Except for the direct connection between medicines and symptoms, we also consider multi-hop relations, such as ("medicine", "cure", "disease") and ("disease", "has symptoms", "symptoms"). As for the attributes of medicine, we use two kinds of attributes: sub-molecular structure and category. The sub-molecular structure information is extracted by RDKit[3] from its SMILES.

— **TCM**. TCM [61] contains a set of prescriptions, which contains a pair of medicine set mc and symptom set sc. For knowledge graph embedding task, we construct the knowledge graph based on the prescription set. The entity $\mathcal{E}$ is the combination of symptom set and medicine set. In addition, we define four different relations: "HerbSym", "SymHerb", which describe the interaction between the pair of medicine set and symptom set, and "HerbHerb", "SymSym", which indicate the interaction within the symptom set or medicine set. As for the attributes of medicine, we use meridians (the medicine affects which part of the body) and categories.

4.2 Evaluation Metrics

As for evaluation metrics, for the knowledge graph embedding task, we use the general metrics for link prediction, including **mean reciprocal rank** (**MRR**) and Hit@N (N = 1,3,10).

[3]https://www.rdkit.org/.

— *MRR* (Mean Reciprocal Rank). MRR takes the average of the reciprocal rank assigned to the true triple.
— $Hit@N(N = 1, 3, 10)$ (Hits over top-N items). $Hits@N$ measures the percentage of cases in which the true triple appears in the top k ranked triples.

To evaluate the performance of medicine recommendation, we adopt the following measurements for each test case (sc, mc):

— $Precision@K$ (Precision calculated over top-K items). The $Precision@K$ score is the metric that calculates the hit ratio in top K position.

$$Precision@K = \frac{|Top(sc, K) \cap mc|}{K}, \tag{13}$$

where $Top(sc, K)$ is the top K recommended medicines with the highest prediction score given sc, and $|Top(sc, K) \cap mc|$ means the size of the intersection of $Top(sc, K)$ and mc.
— $Recall@K$ (Recall calculated over top-K items). The $Recall@K$ score is the metric that calculates the proportion of test cases that recommends the correct items in a top K position.

$$Recall@K = \frac{|Top(sc, K) \cap mc|}{|mc|}, \tag{14}$$

where $|mc|$ is the size of set of medicines in a test prescription.
— $nDCG@K$ (normalized discounted cumulative gain over top-K items). The $nDCG@K$ score measures the ranking quality.

$$nDCG@K = \frac{DCG@K}{IDCG@K}. \tag{15}$$

$DCG@K$ is the discounted cumulative gain over the top K items. The premise of $DCG@K$ is that the ground truth appearing lower in the retrieved results should be penalized as the graded relevance value is reduced logarithmically proportional to the position of the result. That is if the hit medicine ranks higher in the retrieved result, it gains a larger score. The $nDCG@K$ is the $DCG@K$ normalized by the $IDCG@K$, which is the ideal discounted cumulative gain.

We set K to 5,10,20. In the Medicine dataset, the range of Precision@5 is 1∼20% rather than 0∼100%. This is because there is only one recommendation medicine for given symptoms in the training and testing cases. As defined in Equation (13), the precision is the correctly retrieved answer in the top K results, therefore the maximal precision is $\frac{1}{K}$, that is 20% for Precision@5, 10% for Precision@10 and 5% for Precision@20.

4.3 Parameter Setting

In the experiments, we randomly initialize the parameters. We set the learning rate to be 0.0003, the training frequency of training module $t = 1$, the dimension of the embedding $d = 256$, the layer of GNN is set to 2. We employ early stopping with a 20-epoch patience, i.e., we stop training if the validation loss does not decrease for 20 consecutive epochs. For each triple, we randomly sample 100 negative triples for testing. We run the model on a Tesla V100 on CentOS, and the code is implemented with PyTorch. An epoch takes around a minute to train.

4.4 Comparison Methods

To evaluate the performance of our method MedRec, we compare MedRec with the recent competitive state-of-the-art baseline methods including the following methods. For the knowledge graph embedding task

— **TransE** [2]: A method which models relationships by interpreting them as translations operating on the low-dimensional embeddings of entities.
— **TransR** [24]: A method that learns embeddings by first projecting entities from entity space to the corresponding relation space and then building translation between projected entities.
— **RotatE** [40]: A model that defines each relation as a rotation from the source entity to the target entity in the complex vector space.
— **DistMult** [58]: A general neural network framework for multi-relational representation learning.
— **ComplEx** [41]: An approach uses Hermitian dot product to make use of complex valued embeddings.
— **NARS** [63]: A heterogeneous GNN that computes neighbor averaged features for random subsets of relation types, and combines the features for the down streaming task.

Among these methods, TransE and TransR are traditional translation based methods, RotatE and ComplEx incorporates complex valued embeddings, and DistMult is a GNN.

For the medicine recommendation task,

— **Pinsage** [62]: A GCN for web-scale recommender systems that incorporates both graph structure and node feature information.
— **GC-MC** [42]: A graph auto-encoder framework based on differentiable message passing on the bipartite interaction graph.
— **NGCF** [49]:A collaborative filtering framework that exploits the user-item graph structure by propagating embeddings on it.
— **PTM** [61]: A topic modeling group recommendation method which characterizes the generative process of prescription in traditional Chinese medicine theory and further incorporates domain knowledge into the topic model.
— **SafeDrug** [59]: A GNN that captures both global and local information of molecules during the recommendation.
— **MKR** [45]: A multi-task recommendation method that utilizes knowledge graph embedding task to assist the recommendation task.

These baselines cover different model characters. PTM is a traditional medicine recommendation method based on the topic modeling. Pinsage, GC-MC, and NGCF are recommendation methods based on the GNN. MKR incorporates knowledge graph embedding to assist the recommendation task.

4.5 Quantitative Result (RQ1)

4.5.1 Medicine Recommendation Results. Tables 4 and 5 show the results of the recommendation on two datasets. We can see:

(i) Among the GNN-based baselines GC-MC, NGCF, and PinSage, GC-MC performs the worst and NGCF performs the best, because GC-MC only utilizes the first-order neighbors while NGCF uses multiple graph convolution layers and can learn more information. In addition, NGCF is capable of discovering more complex user behavior patterns and producing better recommendations.
(ii) Our proposed model outperforms SafeDrug, which could be due to introduction of medicine attribute graph in modeling.
(iii) Our proposed method outperforms all the baselines, which shows that a multi-task framework can learn a more flexible and expressive representation for medicine recommendation.

Table 4. Recommendation Results on Medicine Dataset

Method	GC-MC	PTM	MKR	NGCF	PinSage	SafeDrug	MedRec
Precision@5	0.0119	0.0401	0.0103	0.0265	0.0235	0.0582	**0.0650**
Precision@10	0.0104	0.0334	0.0113	0.0236	0.0194	0.0350	**0.0500**
Precision@20	0.0080	0.0240	0.0088	0.0187	0.0157	0.0193	**0.0350**
Recall@5	0.0594	0.2005	0.0516	0.1324	0.1174	0.2910	**0.3251**
Recall@10	0.1040	0.3442	0.1129	0.2359	0.1936	0.3499	**0.4999**
Recall@20	0.1606	0.4794	0.1759	0.3741	0.3133	0.3854	**0.7008**
nDCG@5	0.0380	0.1533	0.0302	0.0838	0.0709	0.1946	**0.2035**
nCDG@10	0.0524	0.1989	0.0498	0.1171	0.0945	0.2140	**0.2599**
nDCG@20	0.0666	0.2362	0.0655	0.1517	0.1254	0.2211	**0.3108**

The best result is highlighted in bold.

Table 5. Recommendation Results on TCM Dataset

	GC-MC	PTM	MKR	NGCF	PinSage	SafeDrug	MedRec
Precision@5	0.0371	0.2345	0.2325	0.2350	0.1784	0.2485	**0.2667**
Precision@10	0.0284	0.1884	0.1894	0.1878	0.1395	0.1944	**0.2111**
Precision@20	0.0212	0.1365	0.1430	0.1440	0.1062	0.1437	**0.1560**
Recall@5	0.0266	0.1677	0.1735	0.1613	0.1232	0.1767	**0.1918**
Recall@10	0.0403	0.2712	0.2772	0.2553	0.1930	0.2709	**0.3002**
Recall@20	0.0593	0.3863	0.4198	0.3898	0.2920	0.3972	**0.4375**
nDCG@5	0.0438	0.0773	0.2676	0.2724	0.2072	0.2850	**0.3079**
nDCG@10	0.0430	0.0989	0.2781	0.2732	0.2064	0.2848	**0.3114**
nDCG@20	0.0497	0.1166	0.3332	0.3234	0.2443	0.3304	**0.3618**

The best result is highlighted in bold.

Table 6. Knowledge Graph Embedding Results on Medicine Dataset

	MRR	Hits@1	Hits@3	Hits@10
TransE	0.4147	0.3072	0.4637	0.6126
TransR	0.4487	0.3216	0.5145	0.6861
DistMult	0.4217	0.3217	0.4666	0.6008
RotatE	0.4709	0.3468	0.5331	0.7082
ComplEx	0.4497	0.3293	0.5140	0.6713
NARS	0.4843	0.3521	0.5452	0.7024
MedRec	**0.5137**	**0.3992**	**0.5605**	**0.7218**

The best result is highlighted in bold.

This observation positively answers the first **research question** (**RQ**1) and verifies the effectiveness of our proposed architecture.

4.5.2 Knowledge Graph Embedding Results. Although the goal of MedRec is to utilize knowledge information to assist medicine recommendation, it is still interesting to determine whether the recommendation task can benefit knowledge graph embedding task.

Tables 6 and 7 show the results of knowledge graph embedding on two datasets. We can see:

(i) Compared with the translation distance-based and semantic matching baselines, our method gains the highest MRR and Hits@N(N = 1,3,10) at most cases, which shows that the graph

Table 7. Knowledge Graph Embedding Results on TCM Dataset

	MRR	Hits@1	Hits@3	Hits@10
TransE	0.5654	0.4129	0.6502	0.8853
TransR	0.5635	0.4031	0.6542	0.9019
DistMult	0.6273	0.4925	0.7082	0.8960
RotatE	**0.6405**	0.5010	**0.7318**	0.9124
ComplEx	0.6211	0.4891	0.6991	0.8846
NARS	0.6185	0.4701	0.7164	0.9060
MedRec	0.6402	**0.5030**	0.7204	**0.9261**

The best result is highlighted in bold.

convolution layers can aggregate the neighborhood's information to generate better knowledge graph embedding.

(ii) Compared with the heterogeneous graph network based method NARS, our method also shows higher MRR and Hits@N(N = 1,3,10), which shows that our knowledge graph can better handle heterogeneous information in the knowledge graph.

(iii) Our model gains more improvement compared with RotatE on Medicine dataset rather than TCM dataset. It is probably because that Medicine dataset contains more relations than TCM dataset, and the relation-aware GNN can capture more relation-specific information.

In addition, the model also benefits from the multi-task training framework and the proximity constraint from the recommendation is useful.

4.6 Ablation Study (RQ2)

The contribution of the individual components of MedRec to its functionality is discussed in this subsection. We compare MedRec with its four variants:

— **w/o KG** is a variant that trains the recommendation module with Equation (12) only. We pretrain the parameters of knowledge graph embedding module and fix them during training the recommendation module.
— **w/o Rec** is a variant that trains the knowledge graph embedding module with Equation (3) only.
— **w/o Attr** is a variant that removes attribute information, where $e_m = e_m^{kg}$.
— **App** is a variant that directly appends the attribute information to medicine representation.
— **Sim** is a variant that uses graph similarity to learn the medicine representation.
— **HeteG** is a variant that encodes the attribute and knowledge graph in a unified graph.

From Table 8, we observe that MedRec outperforms w/o Attr, which shows that the attribute helps maintain the rich interaction between medicines and benefits the medicine recommendation. Furthermore, MedRec outperforms w/o KG. This shows that our model can benefit from the multi-task training framework. Iterately trains two modules is better than train them one by one separately. This is because knowledge graph embedding task and medicine recommendation task are not only similar but also highly correlated. The results of HeteG show that learning the feature of attributes separately can better capture the relationship within the medicines, which can make up the incompleteness of the knowledge graph. The results of "App" show that learning the feature of attributes using a GNN can better capture the relationship within the medicines. The proposed method outperforms "Sim", which demonstrates that learning over fine-grained functional groups could be more effective than learning over molecular graphs in the context of this task.

Table 8. The Results of Variants

	Precision@5	Recall@5	nDCG@5	MRR	Hits@3
w/o KG	0.0587	0.2937	0.1783	—	—
w/o Rec	—	—	—	0.4869	0.5299
w/o Attr	0.0335	0.1677	0.1078	0.4272	0.4766
App	0.0519	0.2596	0.1603	0.3861	0.4176
Sim	0.0404	0.2018	0.1302	0.3719	0.4288
HeteG	0.0543	0.2714	0.1748	0.4936	0.5374
MedRec	**0.0650**	**0.3251**	**0.2035**	**0.5137**	**0.5605**

The best result is highlighted in bold.

4.7 Parameters Analysis (RQ3)

In this part, we will discuss how the hyper-parameters influence the result in detail.

— Impact of layer number L. The depth of the network can influence the performance, we tune the number of GCN layers. Figure 5(a)–(c) shows the experimental results according to various layers of the GNN. The performance increases when the layer number is two, because it can encode more useful information with more layers. When increasing the layer number to three, there is a performance drop. The reason may be overfitting caused by large propagation depth.

— Impact of recommendation system training frequency t. We investigate the influence of parameters t by varying t from 1 to 5. The results are presented in Figure 5(d)–(f). We observe that MedRec achieves the best performance when $t = 4$. This is because too small of a training frequency of knowledge graph embedding model, result in limited the transferred knowledge from knowledge graph, while too large of a training frequency of knowledge graph embedding may mislead the objective function of recommendation task.

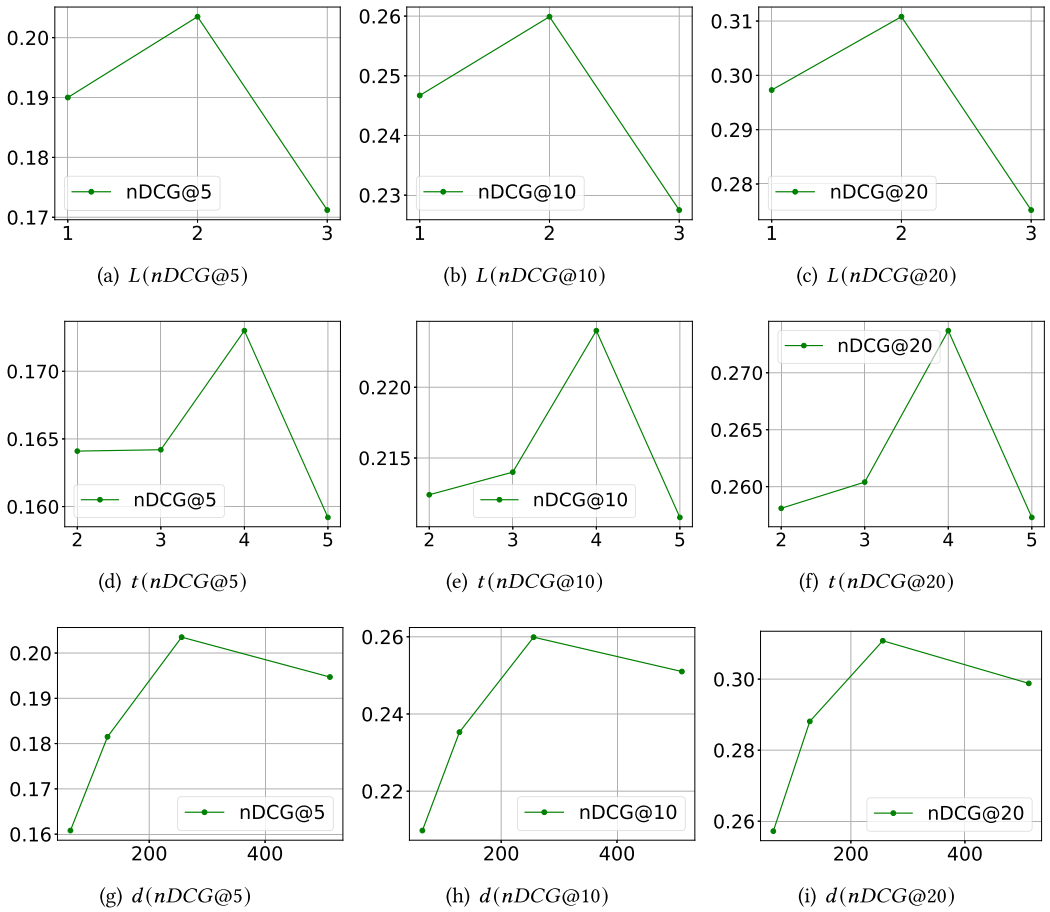
— Impact of embedding dimension d. To explore whether our proposed model can benefit from a larger number of embedding dimension, we investigate the influence of the parameter d. From Figure 5(g)–(i), we find that the performance is initially improved with the increase of dimension because it can encode more information with more bits. However, the performance drops when the dimension is 512, as too large dimensions may introduce noise and make the network hard to be trained.

4.8 Case Study (RQ4)

To show the recommendation results of our model intuitively, we analyze several cases in the recommendation results.

As shown in Figure 6(a), the symptom is “renal colic”, and the recommended medicine is “atropine”, and our model can recommend atropine precisely. In addition, the other top 4 recommended medicines from our model are all anodyne, which have the potential to cure renal colic. Our model can find these medicines because we incorporate the attribute of the medicines, and they all have the attribute anodyne category. Furthermore, scopolamine is listed in the recommended treatment of the Wikipedia page of renal colic. Even though we do not have the knowledge that scopolamine is helpful for “renal colic”, we give this recommendation, which shows that our method can make up the missing knowledge of our knowledge graph.

Another example in Figure 6(c) shows that our method can retrieve more relevant results. For the symptom delusion, the top 5 retrieved medicines from MedRec are antidepressant medication, which is the same category with the recommended “sertraline”. On the other hand, the results



(a) $L(nDCG@5)$ (b) $L(nDCG@10)$ (c) $L(nDCG@20)$

(d) $t(nDCG@5)$ (e) $t(nDCG@10)$ (f) $t(nDCG@20)$

(g) $d(nDCG@5)$ (h) $d(nDCG@10)$ (i) $d(nDCG@20)$

Fig. 5. Performance of different parameters.

Table 9. Performance on Medicine Dataset

Method	NGCF	MKR	GC-MC	ours
Time(s)	27.7	5.63	39	13.4

recommended by the top 5 from w/o Attr are not so accurate. For example, "clonazepam" is a medication used to prevent and treat seizures, panic disorder, and the movement disorder known as akathisiaa medication used to prevent and treat seizures, panic disorder, and the movement disorder known as akathisia and "aripiprazole" is primarily used in the treatment of schizophrenia and bipolar disorder. Examples in Figures 6(b) and 6(d)–(f) also show that our method can recommend the medicines precisely.

4.9 Performance Test

To valid our method's prediction performance, we have run performance test on Medicine dataset. The results are in Table 9. We run the test on Tesla V100 10 times and take the average. For 4,395 test cases in the test set, our method takes 13.4 seconds to generate the recommendation results, which shows the efficiency of prediction and can meet the need in real-world scenario.

Symptom: Renal colic（肾绞痛）
GT：Atropine（阿托品）

Top 5 results from MedRec	Top 5 results from w/o Attr
Atropine（阿托品）	Furosemide（呋塞米）
Pethidine（哌替啶）	Metronidazole（甲硝锉）
Clonidine（可乐定）	Mannitol（甘露醇）
Scopolamine（东莨菪碱）	Atropine（阿托品）
Morphine（吗啡）	Dexamethasone（地塞米松）

(a)

Symptom: Vocal fremitus（语音震颤）
GT：Metronidazole（甲硝锉）

Top 5 results from MedRec	Top 5 results from w/o Attr
Salbutamol（沙丁胺醇）	Penicillin（青霉素）
Metronidazole（甲硝唑）	Metronidazole（布洛芬）
Domperidone（多潘立酮）	Erythromycin（红霉素）
Clindamycin （克林霉素）	Dexamethasone（地塞米松）
Theophylline（茶碱）	Metronidazole（甲硝唑）

(b)

Symptom: Delusion（幻觉妄想）
GT：Sertraline（舍曲林）

Top 5 results from MedRec	Top 5 results from w/o Attr
Sertraline（舍曲林）	Clonazepam（氯硝西泮）
Fluvoxamine（氟伏沙明）	Aripiprazole（阿立哌唑）
Fluoxetine（氟西汀）	Amitriptyline（阿米替林）
Amitriptyline（阿米替林）	Tandospirone（坦度螺酮）
Alprazolam（阿普唑仑）	Sertraline（舍曲林）

(c)

Symptom: Watery stool（水样便）
GT：Lansoprazole（兰索拉唑）

Top 5 results from MedRec	Top 5 results from w/o Attr
Esomeprazole（埃索美拉唑）	Metronidazole（甲硝唑）
Rabeprazole（雷贝拉唑）	Amoxicillin（阿莫西林）
Lansoprazole（兰索拉唑）	Omeprazole（奥美拉唑）
Omeprazole（奥美拉唑）	Lansoprazole（兰索拉唑）
Loperamide（洛哌丁胺）	Ranitidine（雷尼替丁）

(d)

Symptom: Hypoxia（缺氧）
GT：Diazepam（地西泮）

Top 5 results from MedRec	Top 5 results from w/o Attr
Phenobarbitone（苯巴比妥）	Furosemide（呋塞米）
Diazepam（地西泮）	Mannitol（甘露醇）
Paracetamol（对乙酰氨基酚）	Dexamethasone（地塞米松）
Tapazole（甲巯咪唑）	Ibuprofen（布洛芬）
Lorazepam（劳拉西泮）	Diazepam（地西泮）

(e)

Symptom: Dry cough （干咳）
GT：Streptomycin（链霉素）

Top 5 results from MedRec	Top 5 results from w/o Attr
Streptomycin（链霉素）	Penicillin（青霉素）
Rifampicin（利福平）	Rifampicin（利福平）
Isoniazid（异烟肼）	Erythromycin（红霉素）
Dextromethorphan（右美沙芬）	Streptomycin（链霉素）
Penicillin（青霉素）	Ibuprofen（布洛芬）

(f)

Fig. 6. Six examples of recommendation results given the symptoms.

5 CONCLUSIONS

In this article, we explore the medicine recommendation problem, which predicts a set of medicines given a set of symptoms. To alleviate the data sparse problem, we propose MedRec which introduces a medical knowledge graph and a medicine attribute graph. The medical knowledge graph mines the interrelationship between medicine, symptoms, diseases, and physical examinations; the medicine attribute graph mines the inner relationship between medicines. Through these two graphs, we manage to learn the embedding representations of symptoms and medicines which are used for medicine recommendation. To unify the embedding learning and recommendation tasks, we propose a multi-task framework in training and find that these two tasks are highly correlated and can benefit each other. We prove the advantages of MedRec over state-of-art approaches on two datasets, one public and one private. We also illustrate the effectiveness of MedRec with a case study.

Our research leaves room for future work in several directions. One interesting direction is to incorporate the personal information of the patient. Patient electronic health records capture the comprehensive medical histories of patients and can provide personal information for recommendation. In addition, it will be interesting and challenging to extend our framework to explore other paradigms in recommendation systems beyond medicine recommendation, such as treatment recommendation.

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